LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	26
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Side slip sensing and turn co-ordination, location of sensing elements in the A/C and explanation of inertial attitude references.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Side slip sensing and turn co-ordination, location of sensing elements in the A/C and explanation of inertial attitude references.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS sensing element locations on aircraft diagrams; Instructor supervision during diagram analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Where are the side slip sensing elements typically located?
(c) Define inertial attitude references.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Side slip sensing mechanism • Turn co-ordination principles • Sensing element locations in A/C • Inertial attitude reference explanation	L & Ds	PPT, IB	15 min
(b)	• AFCS operational logic • Error detection via inertial refs • Turn coordination feedback loops	L & Ds	PPT, PDF	20 min
(c)	• Integration of side slip data • Inertial reference stability • AFCS output generation	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how side slip sensing contributes to turn coordination.
(b) Describe the role of inertial attitude references in AFCS function.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of side slip sensing and inertial attitude references discussed in this lesson will be utilized in the study of inductive elements like EI bar and Synchro in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Inductive element like EI bar and Synchro as an error detection element in FCS, torque synchro, control synchro, differential and Resolver synchro.	Refer to the master syllabus reference list	LP 27

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min